

# Programowanie gier

## Historia i klasyfikacja gier

Marek Kopel

wideo z wykładu:

[https://pwr-edu.zoom.us/rec/share/B94ly458iaO\\_-NseGOtBGImyTNsuZ6yMKHSZ18IfvTz02cFV-BwpjjdGDeWRphua.SyiC6W-txLS1glQt?startTime=1650885370000](https://pwr-edu.zoom.us/rec/share/B94ly458iaO_-NseGOtBGImyTNsuZ6yMKHSZ18IfvTz02cFV-BwpjjdGDeWRphua.SyiC6W-txLS1glQt?startTime=1650885370000)

# History of video games outline

Early history (1948–1972)

A new industry

Early arcade video games (1972–1978)

First generation of home consoles and the *Pong* clones (1972–1978)

Mainframe computer games (1971–1979)

Golden age

Golden age of arcade video games (1978–1982)

Second generation consoles (1976–1982)

Early home computer games (1976–1982)

1980s

Gaming computers

Early online gaming

Handheld LCD games

Video game crash of 1983

Third generation consoles (1983–1995) (8-bit)

Fourth generation consoles (1987–2004) (16-bit)

1990s

Resurgence and decline of arcades

Handhelds come of age

PC gaming

Fifth generation consoles (1993–2005) (32- and 64-bit)

Transition to 3D and CDs

Mobile phone gaming

2000s

Sixth generation consoles (1998–2013)

Return of alternative controllers

Online gaming rises to prominence

Mobile games

Seventh generation consoles (2005–present)

Increases in development budgets

Rise of casual PC games

Cloud computing comes to games

2010s

Eighth generation consoles (2012–present)

2020s

Ninth generation (2020–present)



toster  
router  
fridge

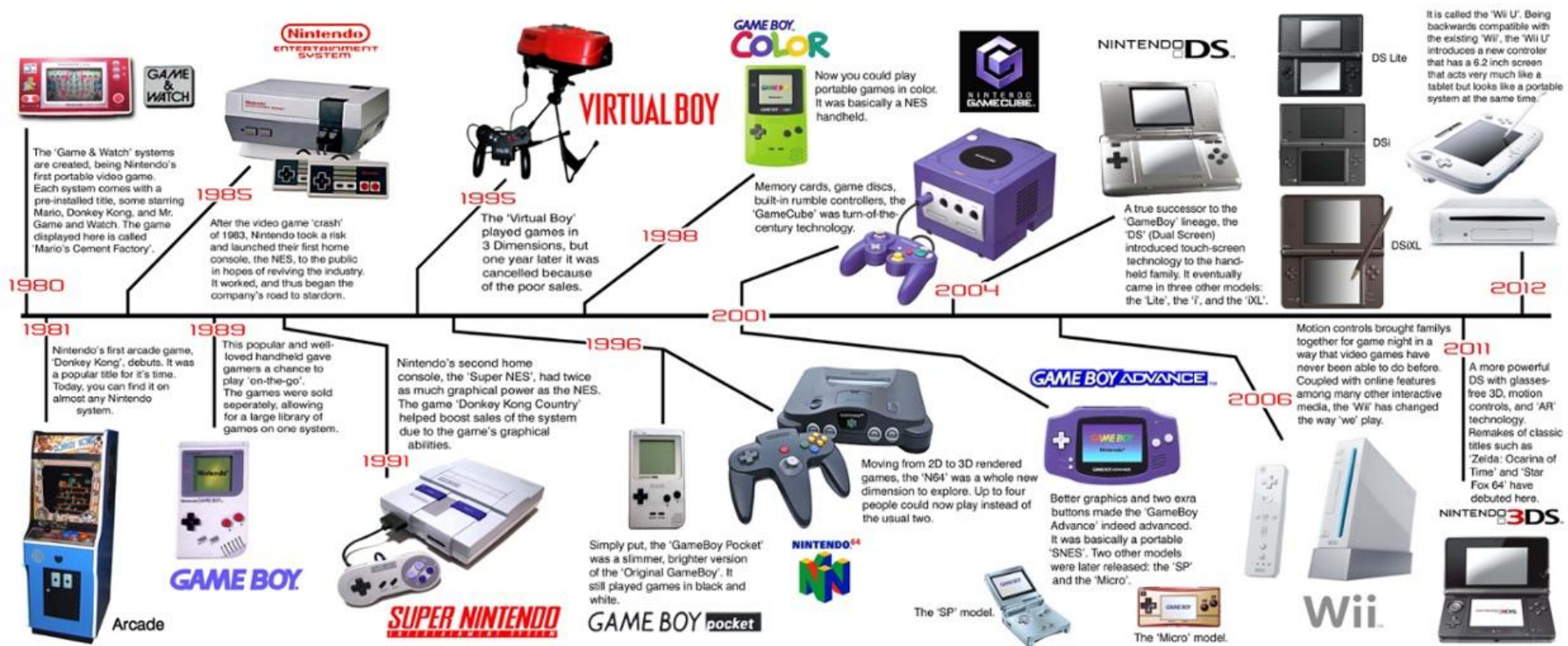


# Xbox (2001- ) and PlayStation (1994- ) evolution





# Nintendo timeline





1985



1991



1996



2001



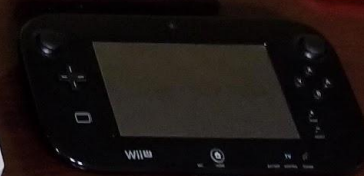
2006



2012



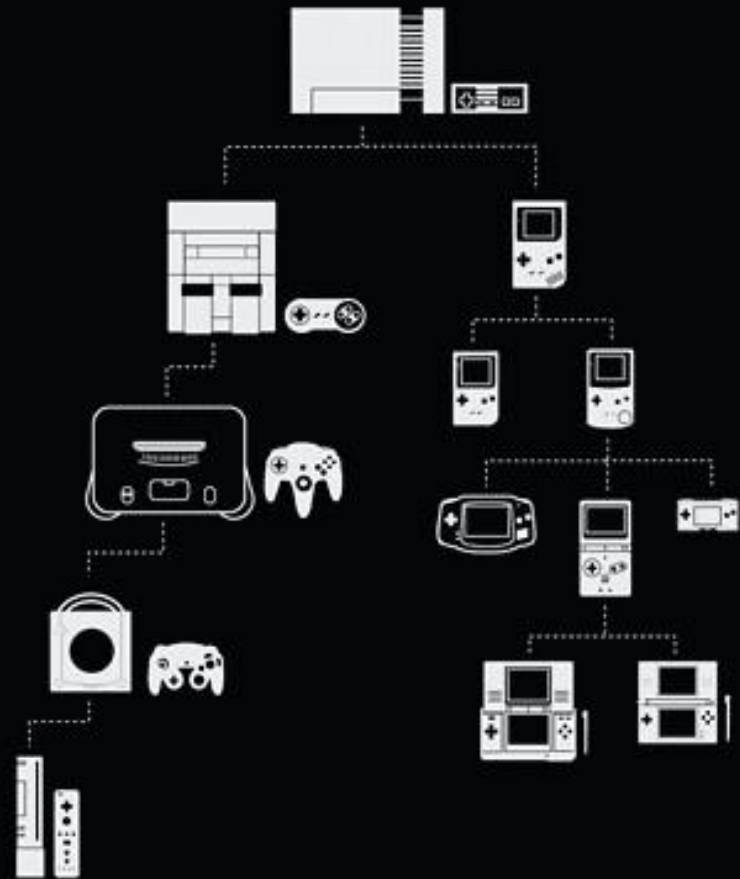
2017



# HANDHELD EVOLUTION

## NINTENDO

1980-PRESENT



NerdyShirts.com



1980



1989



1998



2001



2004



2006



2011

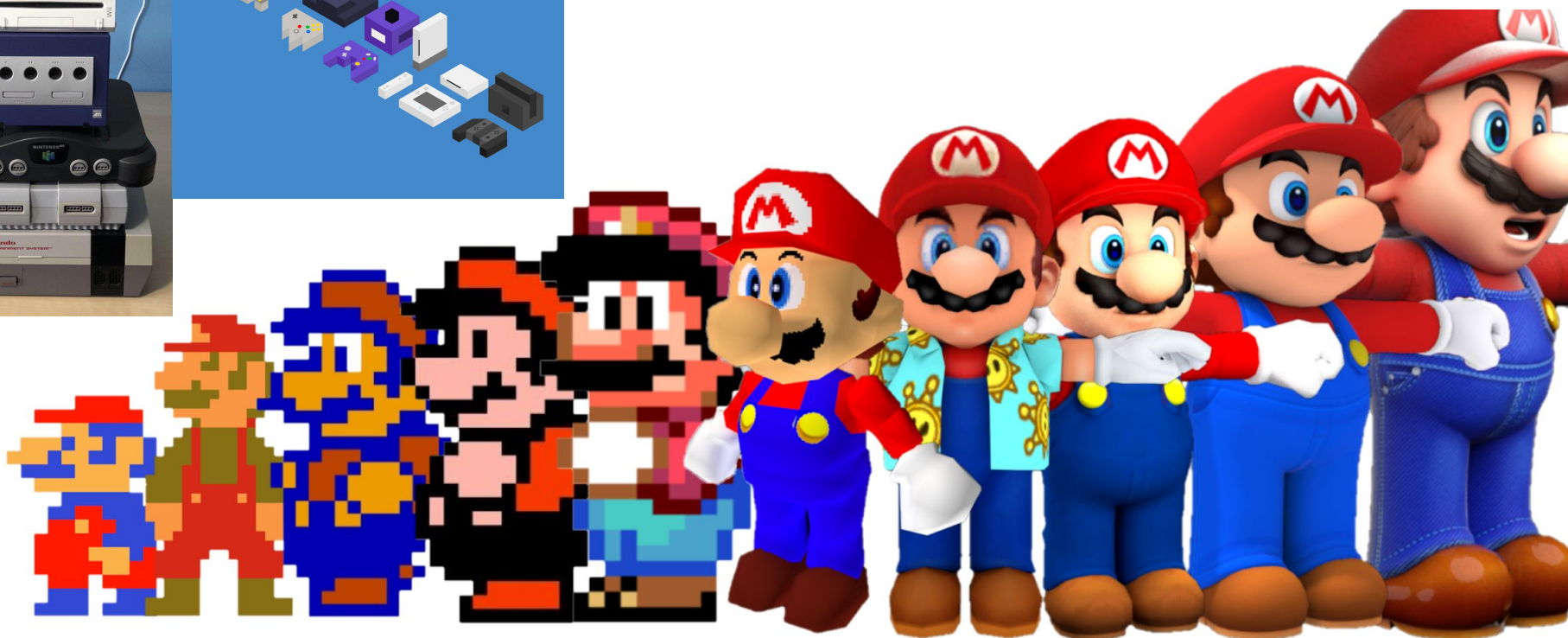
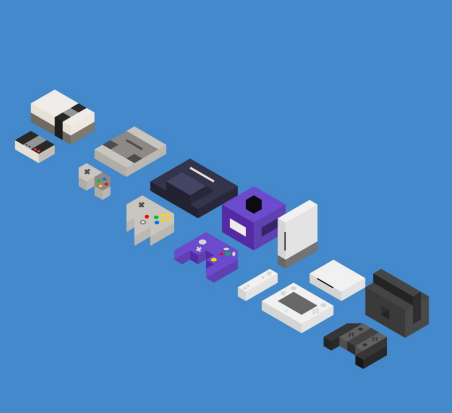


2015



2017





## 2001-2007

2017





# Console platforms





## Video games (graphics) evolution



# Lara Croft (1996-) Tomb Raider series



# Video game genre

[https://en.wikipedia.org/wiki/Video\\_game\\_genre](https://en.wikipedia.org/wiki/Video_game_genre)

A video game genre is a **classification** assigned to a video game **based primarily on its gameplay (type of interaction)** rather than visual or narrative features. A video game genre is normally **defined by a set of gameplay challenges** considered **independently of setting** or game-world content, unlike works of fiction that are expressed through other media, such as films or books.

For example, a **shooter game** is still a shooter game, **regardless of where or when** it takes place.

As with nearly all varieties of genre classification, the matter of a specific game's genre is open to **subjective interpretation**. An individual game may **belong to several genres at once**.



# [https://en.wikipedia.org/wiki/Template:Video\\_game\\_genre](https://en.wikipedia.org/wiki/Template:Video_game_genre)

| V · T · E                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Video game genres | [hide] |
|-----------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|--------|
| List of video game genres · Lists of video games by genre |                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                   |        |
| <b>Action</b>                                             | Beat 'em up (hack and slash) · Fighting · Maze ( <i>Pac-Man</i> clone) · Platform · Shooter (first-person · third-person · light gun · shoot 'em up · arena · hero · tactical) · Survival (battle royale)                                                                                                                                                                                                                                                       |                   |        |
| <b>Action-adventure</b>                                   | <i>Grand Theft Auto</i> clone · Immersive sim · Metroidvania · Stealth · Survival horror                                                                                                                                                                                                                                                                                                                                                                        |                   |        |
| <b>Adventure</b>                                          | Escape the room · Interactive fiction · Interactive film · Visual novel                                                                                                                                                                                                                                                                                                                                                                                         |                   |        |
| <b>MMO</b>                                                | MMOFPS · MMORPG · MMORTS · MUD                                                                                                                                                                                                                                                                                                                                                                                                                                  |                   |        |
| <b>Role-playing</b>                                       | Action role-playing · Dungeon crawl · Roguelike · Tactical role-playing                                                                                                                                                                                                                                                                                                                                                                                         |                   |        |
| <b>Simulation</b>                                         | Construction and management (business · city-building · government) · Falling-sand · Life simulation (dating sim · digital pet · God · social simulation) · Sports                                                                                                                                                                                                                                                                                              |                   |        |
| <b>Strategy</b>                                           | 4X · Auto battler · Multiplayer online battle arena · Real-time strategy (time management) · Real-time tactics · Tower defense · Turn-based strategy · Turn-based tactics (artillery) · Wargame (grand strategy wargame)                                                                                                                                                                                                                                        |                   |        |
| <b>Vehicle simulation</b>                                 | Flight simulator (amateur · combat · space) · Racing (kart racing · sim racing) · Submarine simulator · Train simulator · Vehicular combat                                                                                                                                                                                                                                                                                                                      |                   |        |
| <b>Other genres</b>                                       | Deck-building (roguelike deck-building) · Digital tabletop (digital collectible card) · <i>Eroge</i> · Exergame · Horror · Incremental · Music (rhythm) · Non-game · Party · Programming · Puzzle (sokoban · tile-matching) · Typing                                                                                                                                                                                                                            |                   |        |
| <b>Related concepts</b>                                   | AAA game · Advertising (advergame) · Arcade game · Art game · Audio game · Casual game · Christian game · Crossover game · Educational game · FMV · Gamification · Indie game · Multiplayer video game · Nonlinear gameplay (open world · sandbox game) · Nonviolent video game · Online game (browser game · online gambling · social-network game) · Pervasive game · Serious game · Toys-to-life · Twitch gameplay · Virtual reality game · Video game clone |                   |        |



# Classifications

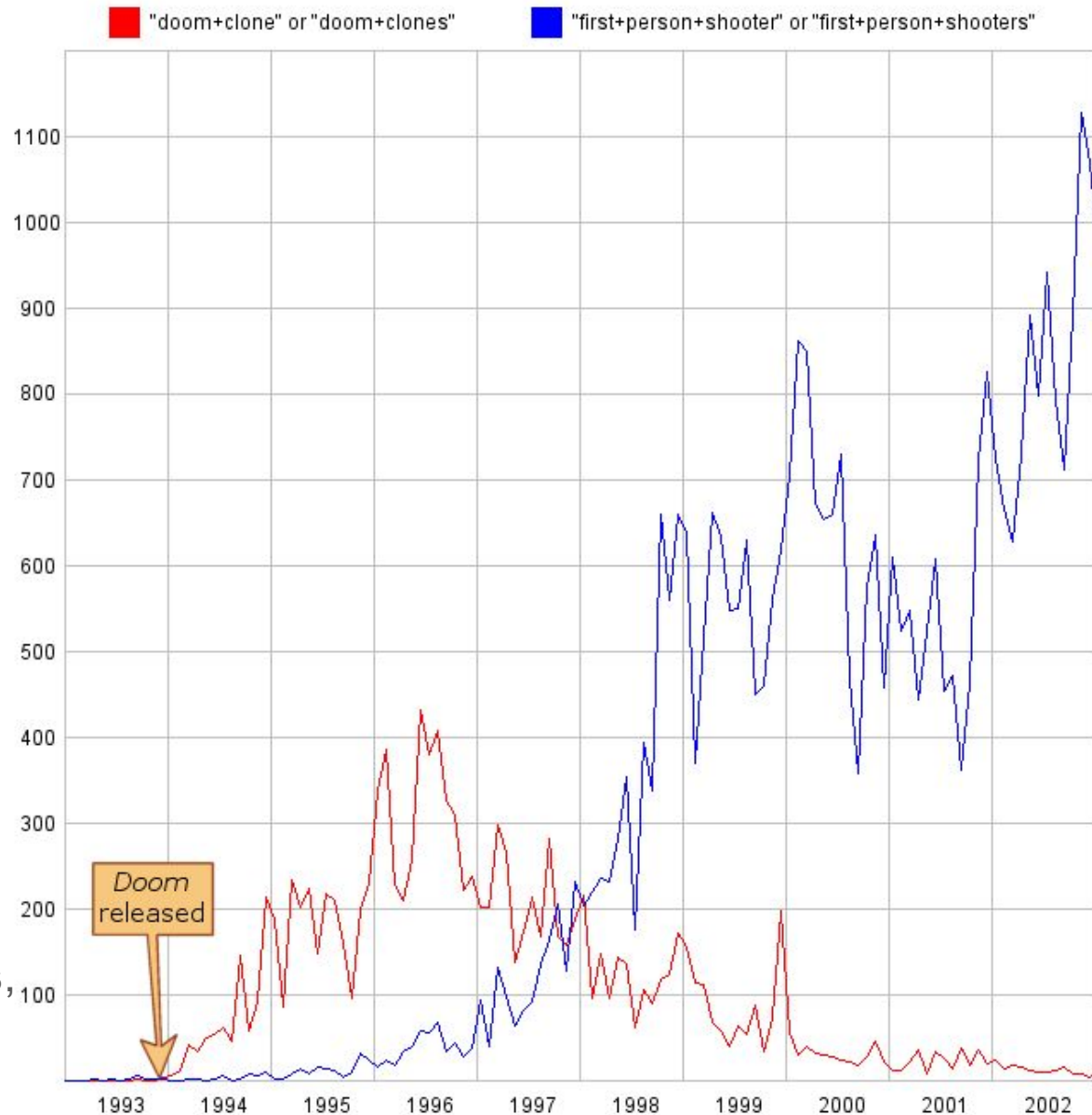
“Doom clone” FPS

“Grand Theft Auto clone”  
action-adventure

“metroidvania”  
action-adventure  
from Metroid and Castlevania

“roguelike” from Rogue  
RPG subgenre  
characterized by  
**dungeon crawl** through  
**procedurally generated levels**,  
**turn-based gameplay**,  
**grid-based movement**,  
and **permanent death**

Usenet posts per month containing the phrase "doom clone", vs "first person shooter"



Data obtained from Google Groups (<http://google.com/grphp>)

# Also by

- By **platform**: generally to the nature of the computer hardware that the game is played on and not the specific branding. This would include game genres like [mobile games](#) for [smartphone](#), [tablet computers](#), or other similar portable devices; and [browser games](#) that can be played in a [web browser](#). Identification of the type of hardware a game is played on implied certain limits to the type of gameplay that is available; a mobile game will typically lack as much action compared to a game playable on a home console or computer due to limitations on player input.
- By **mode**: referring to whether a game is [single player](#), [multiplayer](#), or variations on that, including [massively multiplayer online](#) (MMO) games, [cooperative games](#), [player versus environment](#) (PvE) or [player versus player](#) (PvP) games, and so forth.
- By **narrative**: Classifying video games by their narrative style, such as science fiction or fantasy, is typically not used within the field, with the key exception of [horror games](#), which broadly cover any game dealing with elements of [horror fiction](#).

| Genre        | Softalk<br>1980-1984 | VGC top 100 |      |      |      |      |      |      |      | ESA  | Sta  |
|--------------|----------------------|-------------|------|------|------|------|------|------|------|------|------|
|              |                      | 1995        | 2000 | 2005 | 2010 | 2015 | 2016 | 2017 | 2018 |      |      |
| Action       | 61                   | 3           | 4    | 12   | 15   | 27   | 25   | 22   | 29   | 22.5 | 26.9 |
| Adventure    | 11                   | 2           | 4    | 7    | 6    | 2    | 1    | 0    | 1    | 7.8  | 7.9  |
| Fighting     |                      | 15          | 10   | 5    | 2    | 5    | 3    | 5    | 5    | 5.8  | 7.8  |
| Platform     |                      | 10          | 7    | 10   | 9    | 4    | 3    | 4    | 9    |      |      |
| Puzzle       |                      | 9           | 2    | 6    | 1    | 0    | 0    | 1    | 1    |      |      |
| Racing       |                      | 6           | 6    | 13   | 8    | 5    | 4    | 6    | 6    | 3.3  | 5.8  |
| Role-playing | 18                   | 18          | 25   | 7    | 16   | 12   | 15   | 17   | 12   | 12.9 | 11.3 |
| Shooter      |                      | 11          | 1    | 8    | 14   | 22   | 24   | 19   | 13   | 27.5 | 20.9 |
| Simulation   |                      | 6           | 7    | 5    | 0    | 4    | 4    | 0    | 2    |      |      |
| Sports       |                      | 9           | 19   | 17   | 16   | 12   | 13   | 15   | 13   | 11.7 | 11.1 |
| Strategy     | 10                   | 7           | 8    | 3    | 1    | 0    | 0    | 2    | 1    | 4.3  | 3.7  |



# Coding?

tylko **ZAJAWKA**  
- całość na  
**Gry Komputerowe**  
II stopień,  
spec. PSI

**What was the game Pong's original coding? What language did it use?**

<https://www.quora.com/What-was-the-game-Pongs-original-coding-What-language-did-it-use>

“The first Pong was like that self-assembly thing. Just a bunch of hardware with processors. And processors back then used **machinecode** that could be generated from Assembly. C could have been used but **C was barely born** back then so Assembly or raw **machinecode** would be the more logical choices.”

# Coding? (cont.)

## What programming language was Super Mario Bros. written in?

<https://gamedev.stackexchange.com/questions/6145/what-programming-language-was-super-mario-bros-written-in>

“Almost all NES games were hand-written in **6502 assembly**, the same as used in the **Commodore 64**, the **Apple ][e**, etc. The very few which were written in C had a reputation for being terribly slow because the **NES is only 2Mhz and has 2Kb of onboard RAM** (with an 8Kb window for the cartridge to patch more in). Careful assembly designed to take full advantage of the NES's peculiar architecture was much more effective.

Stop and think about that... games like **Super Mario Bros 3** and **Kirby's Adventure** ran on **2Mhz**. Compare how rich they are to **today's games** which **swallow gigahertz and hundreds of megabytes of RAM...**”

## Writing NES Games! With Assembly!!

<https://www.youtube.com/watch?v=kXbMCKMJXXQ>

# Coding? (cont.)

## Assembly guide

[https://en.wikibooks.org/wiki/6502\\_Assembly](https://en.wikibooks.org/wiki/6502_Assembly)

[https://en.wikibooks.org/wiki/NES\\_Programming](https://en.wikibooks.org/wiki/NES_Programming)

[https://en.wikibooks.org/wiki/Super\\_NES\\_Programming](https://en.wikibooks.org/wiki/Super_NES_Programming)

## NES Starting example

<https://github.com/bbbradsmith/NES-ca65-example>

## PC Engine example (Ball, Walker)

<http://www.magicengine.com/>

# Coding? (cont.)

C teenage years, 3D game is born

[https://www.facebook.com/story.php?story\\_fbid=10157949436169725&id=530059724](https://www.facebook.com/story.php?story_fbid=10157949436169725&id=530059724)



# Donkey Kong - “znak czasu”

tylko **ZAJAWKA**  
- całość na  
**Gry Komputerowe**  
II stopień,  
spec. PSI



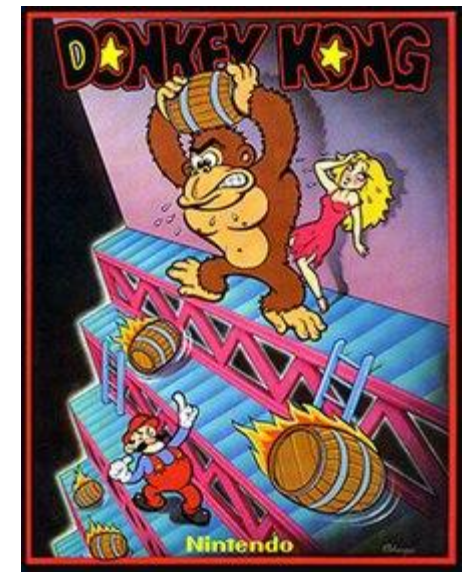
# Story? {START DK digression}

[https://en.wikipedia.org/wiki/Donkey\\_Kong\\_\(video\\_game\)#Plot](https://en.wikipedia.org/wiki/Donkey_Kong_(video_game)#Plot)

Donkey Kong is considered to be the **earliest** video game with a **storyline that visually unfolds** on screen. The eponymous Donkey Kong character is the game's de facto **villain**. The **hero is a carpenter** originally unnamed in the Japanese arcade release, later named **Jumpman** and then **Mario**. Donkey Kong kidnaps Mario's girlfriend, originally known as **Lady**, but later renamed **Pauline**.[...]

Mario, depicted in red overalls and a red cap[...] Graphical limitations and the **low pixel resolution** of the small sprites prompted his design: drawing a mouth with such few pixels is infeasible, so the character **was given a mustache**; the programmers could **not animate hair**, so he got a **cap**; and to **make his arm movements visible**, he needed colored overalls.

The **artwork** used for the cabinets and promotional materials make these **cartoon-like** character designs even more explicit. Pauline, for example, is depicted to be disheveled (like King Kong's Fay Wray) in a torn dress and stiletto heels.



# Story? (cont.)

Donkey Kong is the first example of a **complete narrative told in video game form**, and like 1980's Pac-Man, it **employs cutscenes** to advance its plot.

“Although its origin as a comic strip license [Popeye love triangle - ed.] played a major part, Donkey Kong marked **the first time that the storyline for a video game preceded the game's programming** rather than simply being appended as an afterthought”





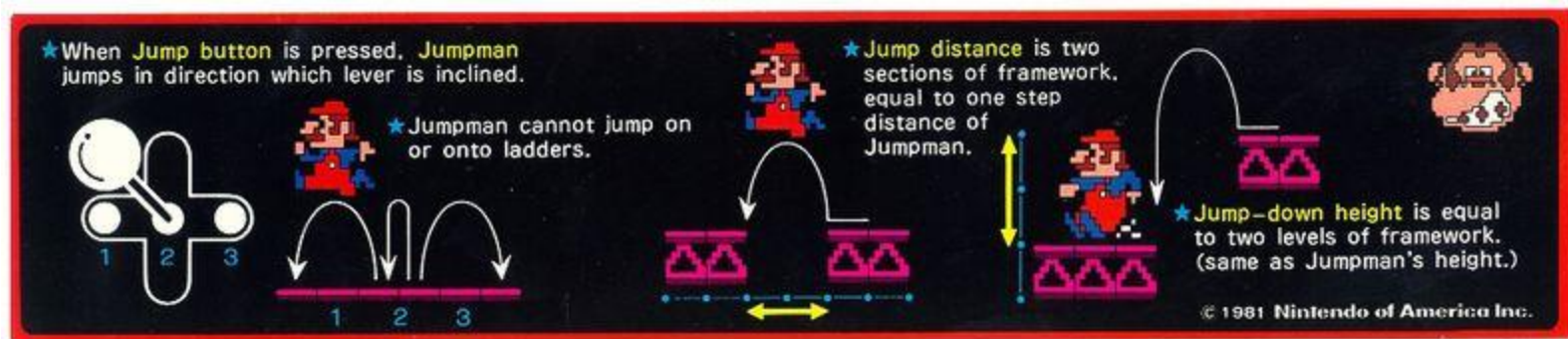
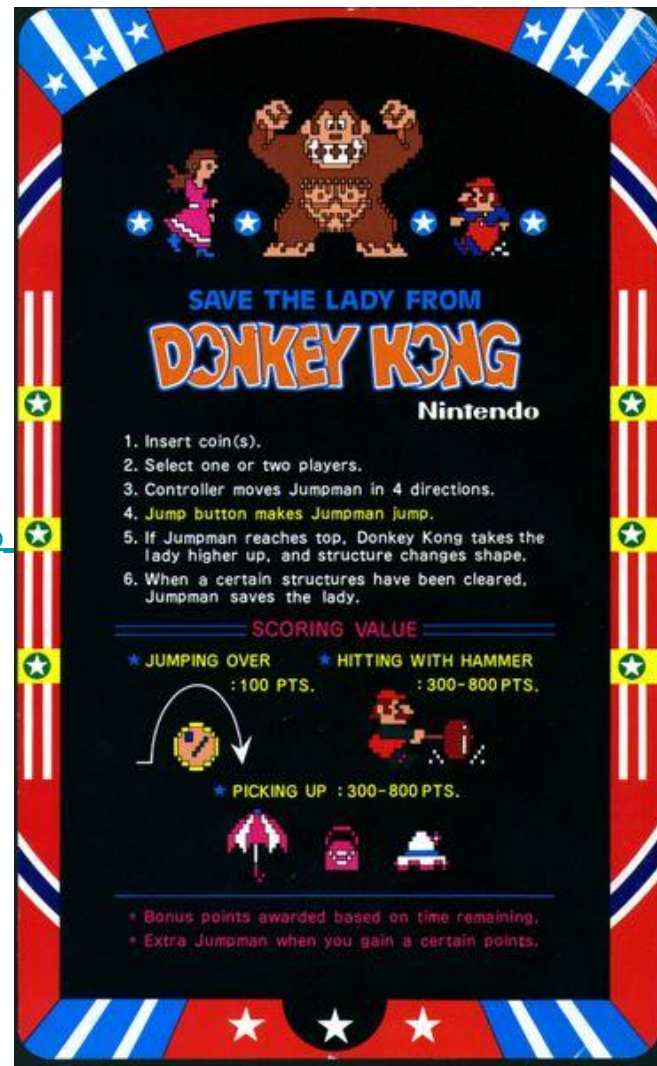
# How player knows the story? Manual?

rewritten

[https://www.mariowiki.com/Donkey\\_Kong\\_\(game\)#Official\\_story\\_quoted\\_from\\_Nintendo](https://www.mariowiki.com/Donkey_Kong_(game)#Official_story_quoted_from_Nintendo)



and for NES (already Mario)



# Development

[https://en.wikipedia.org/wiki/Donkey\\_Kong\\_\(video\\_game\)#Development](https://en.wikipedia.org/wiki/Donkey_Kong_(video_game)#Development)

As of late 1980 to early 1981, Nintendo's efforts to expand to North America had failed, culminating with the attempted export of the otherwise successful Radar Scope. They were left with a large number of **unsold Radar Scope machines**, so company president Hiroshi Yamauchi thought of simply **converting them into something new**. He approached a young **industrial designer named Shigeru Miyamoto**, who had been working for Nintendo since 1977, to see if he could **design such a replacement**. Miyamoto said that he could. Yamauchi appointed Nintendo's head engineer, Gunpei Yokoi, to supervise the project. Nintendo's budget for the development of the game was \$100,000. Some sources also claim that Ikegami Tsushinki was involved in some of the development. They played no role in the game's creation or concept, but were hired to provide "**mechanical programming assistance** to fix the software created by Nintendo".

The circuit board of Radar Scope was restructured for Donkey Kong. The Radar Scope hardware, originally **inspired by the Namco Galaxian** hardware, was designed for a **large number of enemies moving around at high speeds**, which Donkey Kong **does not require**, so the development team **removed unnecessary functions and reduced the scale of the circuit board**. While the gameplay and graphics were reworked for **updated ROM chips**, the existing CPU, sound hardware and monitor were left intact. The character set, scoreboard, upper HUD display, and font are **almost identical to Radar Scope**, with palette differences. The Donkey Kong hardware has the memory capacity for displaying **128 foreground sprites at 16x16 pixels each and 256 background tiles at 8x8 pixels each**. Mario and all moving objects use single sprites, the taller Pauline uses two sprites, and the **larger Donkey Kong uses six**





# Bugs?

DK: “The 22nd level is colloquially known as the **kill screen**, due to an error in the game's programming that kills Mario after a few seconds, effectively ending the game.”

[https://en.wikipedia.org/wiki/Donkey\\_Kong\\_\(video\\_game\)#Gameplay](https://en.wikipedia.org/wiki/Donkey_Kong_(video_game)#Gameplay)

[https://en.wikipedia.org/wiki/Glossary\\_of\\_video\\_game\\_terms#Kill\\_screen](https://en.wikipedia.org/wiki/Glossary_of_video_game_terms#Kill_screen)



*Donkey Kong* has a kill screen caused by an **overflow condition**, where the game timer kills the player before it is possible to beat the level.<sup>[53]</sup> *Ars Technica* calls it the "**second-most famous kill screen of all of gaming**"<sup>[53]</sup> and *Wired* described it as "**mythic**".<sup>[55][56]</sup> This was popularized in the documentary *The King of Kong*.<sup>[53]</sup>

*Pac-Man* has a kill screen on level 256 based on an **integer overflow**;<sup>[46]:48[54]</sup> *Ars Technica* calls this "**one of the most well known accidental endings in gaming**".<sup>[53]</sup> The games *Ms. Pac-Man* and *Jr. Pac-Man* also have kill screens.<sup>[53]</sup> **Pac-Man's kill screen was playable**, but rendered in such a way that it was not possible to gather sufficient points to advance.



# Arcade revenue

A quarter (25 cent coin) per game = usually 3 lives

“The player typically receives **three lives** with a bonus awarded for 7,000 points, although this can be modified via the game's built in DIP switches.”

[https://en.wikipedia.org/wiki/Donkey\\_Kong\\_\(video\\_game\)#Gameplay](https://en.wikipedia.org/wiki/Donkey_Kong_(video_game)#Gameplay)

“One life is lost whenever Mario (...) lets the bonus counter reach zero”

“Before the advent of cheaper, battery-backed RAM, DIP switches were also often used on arcade games in the 1980s and early 1990s to enter game settings such as difficulty or the **number of credits per coin**.”

[https://en.wikipedia.org/wiki/DIP\\_switch#Applications](https://en.wikipedia.org/wiki/DIP_switch#Applications)

DK cabinet manual - p.5 -

[http://arcarc.xmission.com/PDF\\_Arcade\\_Nintendo/Donkey\\_Kong\\_Operation\\_Manual\\_\(TKG-4-UP\).pdf](http://arcarc.xmission.com/PDF_Arcade_Nintendo/Donkey_Kong_Operation_Manual_(TKG-4-UP).pdf)



more on Arcade DIP

<http://www.classicarcadegaming.com/contests/CAGDCT2015/misc/settings.htm>

# Donkey Kong today

Movie references

Pixels (2015)



live action  
short (2016)



Megamind (2010)



DK became Cranky Kong (?) {END DK digression}

All this fancy  
animation makes me  
sick. Back in my day,  
we'd have two  
frames and enjoy it!



[https://www.mariowiki.com/List\\_of\\_Cranky\\_Kong\\_quotes](https://www.mariowiki.com/List_of_Cranky_Kong_quotes)